

Project Statement

1. Project Title

Personal Expense Tracker

2. Problem Statement

Managing daily expenses manually can make it difficult to keep records organized and calculate how much money is being spent in different categories.

The Personal Expense Tracker is developed to provide a simple command-line application where users can record their expenses, store them in a CSV file, search existing records, calculate spending totals, view monthly summaries, and delete unwanted records.

The project aims to make basic personal expense management easier while demonstrating the use of Python programming concepts.

3. Scope of the Project

The project covers the following activities:

- Recording daily expenses
- Storing expense records in a CSV file
- Viewing all recorded expenses
- Searching expenses by category or description
- Calculating category-wise spending
- Calculating total and average spending
- Generating monthly expense summaries
- Deleting expenses using their IDs
- Validating user input

The project is designed as a local command-line application and does not require an internet connection or an external database.

4. Target Users

The project is intended for:

- Students
- Individuals who want to keep track of daily expenses
- Users who need a simple personal expense record
- Beginners learning Python programming and file handling

5. High-Level Features

The major features of the project are:

1. Add Expense

Allows the user to enter the date, category, description, and amount of an expense.

2. View All Expenses

Displays all saved expense records.

3. Search Expenses

Allows users to search expenses using a category or description.

4. Category Summary

Calculates and displays spending for different expense categories.

5. Total Spending

Displays the total number of expenses, total amount spent, and average expense.

6. Monthly Summary

Displays expenses and total spending for a selected month.

7. Delete Expense

Allows users to remove an expense using its ID.

8. Input Validation

Checks dates, amounts, categories, descriptions, IDs, and menu choices to reduce invalid input.

9. CSV Data Storage

Stores expense information in `expenses.csv` so records can be accessed again when the program is reopened.